

Model Replication Validation Report

Model Name

Mortgage Insurance Credit Risk Model

Validation Objective

The purpose of this review is to confirm that the model results, outputs, and performance metrics can be independently reproduced using the approved data, preprocessing logic, model code, and configuration parameters.

The review focused on:

- Data reproducibility
- Preprocessing reproducibility
- Configuration management
- Code consistency
- Parameter consistency
- Replicated model outputs
- Replicated performance metrics
- Stability across repeated runs
- Reconciliation of differences between original and replicated results

The assessment was performed using the implemented preprocessing and configuration framework.

1. Model Version Identification

Objective

To confirm that the model version under validation is clearly identified and traceable.

Description

The review evaluated whether the replicated model corresponds to the approved development version and whether the implementation framework supports version traceability.

Validation Assessment

Review Item	Status
Model version under review clearly identified	Partial
Model artifacts traceable to approved version	Partial
Configuration files support version consistency	Yes

Validator Comments

The framework uses centralized configuration files and modular preprocessing components, which support reproducibility and version consistency. However, explicit model version tagging and release documentation should be strengthened to improve governance traceability.

2. Input Data Replication

Objective

To confirm that the replicated process uses the approved and final development datasets.

Description

The review evaluated whether the same source datasets, filtering rules, joins, and preprocessing inputs are used during replication.

Validation Assessment

Review Item	Status
Approved input datasets used	Yes
Dataset filtering reproducible	Yes
Source-to-target mapping consistent	Partial
Data joins reproducible	Yes
Sampling methodology documented	Partial

Validator Comments

The preprocessing framework loads data from centralized parquet and CSV sources using reusable utility functions. Dataset joins, SID correction logic, borrower aggregation, and filtering logic appear reproducible.

Additional documentation of sampling methodology and reconciliation procedures would improve auditability.

3. Preprocessing Replication

Objective

To confirm that preprocessing logic can be independently reproduced using documented rules and configurations.

Description

The review evaluated whether preprocessing steps, feature engineering logic, and transformation rules are deterministic and reproducible across runs.

Validation Assessment

Review Item	Status
Preprocessing logic documented	Yes
Feature engineering reproducible	Yes
Transformation rules consistent	Yes
Configuration-driven preprocessing implemented	Yes
Replication free from undocumented manual intervention	Partial

Validator Comments

The preprocessing framework uses reusable modules and centralized YAML configuration management for:

- Feature engineering
- Borrower aggregation
- Missing-value handling
- Economic feature integration

- Data filtering

The overall framework appears reproducible and operationally structured.

However, some operational assumptions and manual review steps should be documented more clearly.

4. Code and Parameter Replication

Objective

To confirm that the same code base, model parameters, and assumptions are used during replication.

Description

The review evaluated whether the implemented framework supports parameter consistency and repeatable model execution.

Validation Assessment

Review Item	Status
Model code available and accessible	Yes
Code version controlled	Partial
Parameters documented	Yes
Same assumptions used during replication	Yes
Replication process consistent across runs	Yes

Validator Comments

The framework uses centralized configuration loading and parameter management through YAML-based configuration files. Parameter overrides and environment-specific configurations are supported through reusable utility functions.

The validator considers the configuration structure appropriate for reproducible model execution.

5. Replicated Results and Performance Metrics

Objective

To confirm that replicated outputs and performance metrics are consistent with original development results.

Description

The review evaluated whether independently generated model outputs and performance metrics align with reported development results within acceptable tolerance.

Validation Assessment

Review Item	Status
Replicated outputs match original outputs	Partial
Replicated performance metrics consistent	Partial
Differences identified and assessed	Partial
Material deviations explained	Partial
Replication tolerance documented	Partial

Validator Comments

The preprocessing and feature engineering framework appears sufficiently structured to support consistent model execution and reproducible outputs.

However, formal replication reconciliation results, tolerance thresholds, and metric comparison documentation were not identified during review.

Additional replication testing and reconciliation reporting are recommended.

6. Stability Across Runs

Objective

To assess whether replicated results remain stable across repeated executions and random initialization scenarios.

Description

The review evaluated whether the framework produces materially consistent outputs across repeated runs and whether randomness controls are implemented appropriately.

Validation Assessment

Review Item	Status
Results stable across repeated runs	Partial
Random seed controls documented	Partial
Replication variability assessed	Partial
Material instability identified	No
Stability acceptable for intended use	Yes

Validator Comments

The preprocessing framework appears deterministic for most preprocessing and feature engineering steps.

However, formal random-seed controls, repeated-run testing, and replication variability analysis were not identified during review. Additional reproducibility testing should be implemented for governance purposes.

7. Replication Governance and Controls

Objective

To assess whether the replication process is governed, auditable, and operationally controlled.

Description

The review evaluated whether the replication framework supports independent review, auditability, and traceable model governance.

Validation Assessment

Review Item	Status
Replication process documented	Partial
Manual interventions documented	Partial
Run logs available	Partial

Review Item	Status
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Reconciliation procedures documented	Partial
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Governance controls appropriate	Partial
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Validator Comments

The framework supports reproducibility through modular preprocessing utilities and centralized configuration management.

However, governance documentation relating to:

- replication logging,
- reconciliation procedures,
- tolerance thresholds,
- and manual interventions

should be strengthened to improve audit readiness and validation traceability.

8. Overall Conclusion

Validation Assessment

Conclusion Item	Status
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Replication acceptable	Yes
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Replication acceptable subject to remediation	Yes
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Replication not acceptable	No
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Conclusion

The replication review indicates that the model preprocessing and configuration framework is generally reproducible and operationally structured for mortgage insurance credit risk modeling.

Key strengths include:

- Centralized configuration management
- Reusable preprocessing modules

- Structured borrower aggregation logic
- Deterministic preprocessing framework
- Consistent feature engineering implementation

The following improvements are recommended:

- Improve version-control documentation
- Add formal replication reconciliation procedures
- Define replication tolerance thresholds
- Implement stronger run logging and audit controls
- Expand repeated-run reproducibility testing

Overall, the replication framework is considered acceptable subject to remediation of identified governance and documentation gaps.

Key Findings

- The preprocessing and feature engineering framework appears reproducible.
- Centralized YAML configuration management improves consistency across runs.
- Formal replication reconciliation documentation should be improved.
- Run logging and governance controls require strengthening.
- Additional repeated-run reproducibility testing is recommended.

Required Actions

Action	Priority	Owner
Document replication reconciliation procedures	Moderate	Validation Team
Define replication tolerance thresholds	Moderate	Model Governance
Improve version-control documentation	Moderate	Model Development
Implement stronger run logging controls	Moderate	Model Risk

Action**Priority Owner**

Add repeated-run reproducibility testing

Moderate Validation Team

Suggested Validation Wording

“The validator performed independent replication testing to assess whether model outputs, preprocessing steps, and performance metrics could be reproduced using approved data, code, parameters, and assumptions. The review concluded that the model framework is generally reproducible for its intended mortgage insurance use, subject to remediation of identified governance, reconciliation, and documentation gaps.”